Eigenvalue

The roots of the characteristic equation of $A_{n\times n}$ are the eigenvalues of A.

Ex- ① $A = \begin{pmatrix} 2 & 1 \\ 3 & 5 \end{pmatrix}$

$$\psi_A(x) = x^2 - 7x + 7 = 0 \Rightarrow x = \frac{7 \pm \sqrt{21}}{2}$$

$\therefore \frac{7+\sqrt{21}}{2}, \frac{7-\sqrt{21}}{2}$ are the eigen values of A.

A root of $\psi_A(x) = 0$ of multiplicity r is said to be an r-fold eigenvalue of A.

Prop: ① The product of the eigen values of $A_{n\times n}$ is $\det(A)$.

② Zero is an eigenvalue of $A_{n\times n}$ iff A is singular

③ The eigenvalues of a diagonal matrix are its diagonal elements.

④ If λ be an r-fold eigenvalue of A of order n, then 0 is an r-fold eigenvalue of the matrix $A - \lambda I$.

⑤ If λ be an eigen value of a non singular matrix $A_{n\times n}$, then λ^{-1} is an eigenvalue of A^{-1}.

⑥ Prove it later
Let $P_{n\times n}$ be a non singular matrix, then $A_{n\times n}$ and $P^{-1}AP$ have the same eigenvalues.

⑦ Sum of all the eigen values of A = trace of A.

Ex-2: $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Then $\psi_A(x) = x^2 + 1 = 0$

Then the eigenvalues are $i, -i$.

Ex-3 : $A = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 2 & -1 \\ 3 & 2 & -2 \end{pmatrix}$

$$\psi_A(x) = \begin{vmatrix} 1-x & -1 & 0 \\ 1 & 2-x & 0 \\ 3 & 2 & -2-x \end{vmatrix} = 0$$

$$\Rightarrow (1-x)(x^2-2) + (1-x) = 0$$

$$\Rightarrow (1-x)(x^2-1) = 0 \Rightarrow x = \underbrace{1, 1, -1}_{\text{eigenvalues of } A}$$

Proof of the properties :

① HW Hints : $\psi_A(x) = c_0 x^n + c_1 x^{n-1} + \cdots + c_n = 0$

The product of the roots of this eqn is $(-1)^n \frac{c_n}{c_0} = (-1)^n \frac{\det(A)}{(-1)^n} = \det(A)$

② HW Hints : $\psi_A(x) = |A - \lambda I| = 0 \Rightarrow \psi_A(0) = |A|$

Now A is singular $\Rightarrow |A| = 0 \Rightarrow \psi_A(0) = 0 \Rightarrow 0$ is a root of $\psi_A(x) = 0 \Rightarrow 0$ is an eigenvalue of A.

Conversely $\psi_A(0) = 0 \Rightarrow \det(A) = 0 \Rightarrow A$ is singular

③ HW

④ ~~Hints~~ : Let $\psi_A(x) = \det(A - xI) = (x-\lambda)^r \phi(x)$, $\det(A-(x+\lambda)I) = x^r \phi(\lambda+x)$ s.t. ϕ where $\phi(x)$ is a polynomial of degree $n-r$ and $\phi(\lambda) \neq 0$.

The characteristic polⁿ. of $A - \lambda I_n$ is $\det(A - \lambda I_n - xI_n)$

$$= \det(A - (\lambda + x) I_n)$$

(3) HW

Hint: Let $A = \text{diag}(d_1, d_2, \ldots, d_n)$

Then $\psi_A(x) = \det(A - xI) = (d_1 - x)(d_2 - x) \cdots (d_n - x) = 0$

$\Rightarrow$ roots of $\psi_A(x)$ are $d_1, d_2, \ldots, d_n$.

(4) Let $\psi_A(x) = \det(A - xI_n) = (x-\lambda)^r \phi(x)$, where $\phi(x)$ is a polⁿ of degree $n-r$ and $\phi(\lambda) \neq 0$

$\therefore \det(A - (x+\lambda) I_n) = x^r \phi(x+\lambda) \longrightarrow (1)$

Now the characteristic polⁿ of $(A - \lambda I_n)$ is

$$\det(A - \lambda I_n - x I_n)$$

$$= \det[A - (\lambda + x) I_n] = x^r \phi(x+\lambda)$$

$$= x^r \mu(x),$$

where $\mu(x) = \phi(x+\lambda)$. Now $\phi(\lambda) \neq 0 \Rightarrow \mu(x) \neq 0$

$\therefore$ 0 is a root of $\det(A - \lambda I - xI)$ with the multiplicity r

(5) A is non-singular $\Rightarrow A^{-1}$ exists and 0 is not an eingenvalue of A.

λ is an eigenvalue of $A \Rightarrow \det(A - \lambda I_n) = 0 \longrightarrow (1)$

Now $\det(A^{-1} - \lambda^{-1} I_n) = \det(A^{-1}(AA^{-1} - \lambda^{-1} A I_n))$

$$= \det(A^{-1}) \det(\lambda^{-1}(\lambda A A^{-1} - A))$$

$$= \det(A^{-1}) (\lambda^{-1})^n \det(\lambda I_n - A)$$

$$= (-1)^n (\lambda^{-1})^n \det(A^{-1}) \det(A - \lambda I_n)$$

$$= 0$$

(6) $\psi_{P^{-1}AP}(x) = \det(P^{-1}AP - xI)$

$= \det[P^{-1}AP - P^{-1}(xI)P] \quad \because P^{-1}IP = I$

$= \det[P^{-1}(A - xI)P]$

$= \det(P^{-1}).\det(A - xI).\det(P)$

$= \det(A - xI)\det(P^{-1})\det(P)$

$= \det(A - xI)\det(P^{-1}P) = \det(A - xI) = \psi_A(x)$

(7) Let $\alpha_1, \alpha_2, \ldots, \alpha_n$ be the roots of the equation

$\psi_A(x) = (-1)^n (x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_n) \rightarrow$ ~~(1)~~

$\Rightarrow c_0 x^n + c_1 x^{n-1} + \cdots + c_n = (-1)^n (x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_n) \rightarrow (1)$

Equating the coefficient of x^{n-1} on the both side we get

$c_1 \left(-(-1)^{n-1} \operatorname{trace}(A)\right) = (-1)^{n+1} \sum_{i=1}^{n} \alpha_i$

$\Rightarrow \operatorname{trace}(A) = \sum_{i=1}^{n} \alpha_i.$

Eigen vector

Let $X = (x_1, x_2, \ldots, x_n)$ be a vector (in a vector space $V_n(F)$ over the field F).

X can be represented as $\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$

Let $A_{n \times n}$ be a matrix (over the field F) and

$AX = Y \Rightarrow Y$ is also a vector like $\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \in V_n(F)$

~~Let θ be any number (i.e. any element in F). Then X and θX are the same vector.~~

Eigen vector $\equiv$ self vector

Let X be a non-zero vector and AX ~~$= \lambda X$~~ produces the same vector with the ~~same~~ different magnitude λ, i.e. $AX = \lambda X$, then X is called an eigenvector of A.

Def: Let A be $n \times n$ matrix (over a field F). A non null ~~such~~ vector X ($\in V_n(F)$) is said to be an eigen-vector of A if there ~~is~~ exists a scalar λ ($\in F$) such that $AX = \lambda X$ holds.

Prop 1: λ is an eigenvalue of a matrix $A_{n\times n}$ iff there exists a non zero vector X such that $AX = \lambda X$.

Sol^n: $\Rightarrow$ Since λ is an eigenvalue of A, we have $|A-\lambda I| = 0$

$\Rightarrow$ the matrix $(A-\lambda I)$ is singular

$\Rightarrow$ there exists a non-zero solution of the system of linear homogeneous eqns $(A-\lambda I)X = 0$

$\Rightarrow$ $AX = \lambda X$, where X is a non-zero vector.

[So for any eigenvalue λ, $\exists$ atleast one eigenvector X. In fact there could be many]

$\Leftarrow$ Let there be a non-zero vector X s.t.

$$AX = \lambda X \text{ for some scalar } \lambda.$$

$$\text{i.e. } (A-\lambda I)X = 0 \longrightarrow (1)$$

Since there exists a non zero solution X of the system of homogeneous eqns (1), $\det(A-\lambda I) = 0$.

Thus λ is an eigenvalue of A.

Prop 2: To an eigen vector of $A_{n\times n}$ there corresponds a unique eigenvalue of A.

Sol^n: Let us assume there be two distinct eigenvalues λ_1, λ_2 of A corresponding to an eigenvector X. Then $AX = \lambda_1 X$ and $AX = \lambda_2 X$

$$\Rightarrow \lambda_1 X = \lambda_2 X$$

$$\Rightarrow (\lambda_1 - \lambda_2) X = 0$$

which is a contradiction since $X \neq 0$ and $(\lambda_1 - \lambda_2) \neq 0$

Ex- $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. The eigenvalues of A are $i, -i$.

If we consider A be a real matrix then A has no eigenvalues since, there is no real vector satifies $AX = iX$ or $AX = -iX$.

If we consider A as a complex matrix then we can obtain the eigenvectors associated with the eigenvalues i and $-i$.

Let $X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ be an eigenvector corresponding to i.

Then $AX = iX \Rightarrow \begin{cases} -ix_1 - x_2 = 0 \\ x_1 - ix_2 = 0 \end{cases}$

$$\Rightarrow (A - iI)X = 0$$

$$\Rightarrow \begin{pmatrix} -i & -1 \\ 1 & -i \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$\Rightarrow \begin{pmatrix} 1 & -i \\ -i & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \qquad R_1 \leftrightarrow R_2$$

$$\Rightarrow \begin{pmatrix} 1 & -i \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \qquad R_2 = R_2 + iR_1$$

$\Rightarrow \quad x_1 - ix_2 = 0 \longrightarrow (1)$ free variable

choose $x_2 = t \in \mathbb{C} \Rightarrow (x_1, x_2) = t(i, 1)$

$\therefore$ The eigenvectors corresponding to i are $t(i, 1)$, where t is any non zero complex number.

Similarly the eig. vec. corresponding to $-i$ are $c(1, i)$, where $0 \neq c \in \mathbb{C}$

$$AX = \lambda X \Rightarrow (A-\lambda I)X = 0$$

Rank of $(A-\lambda I)$

Prop (1) If λ be an r-fold eigenvalue of $A_{n\times n}$ then
$\text{rank}(A-\lambda I) \geq n-r$

(2) If λ be a simple eigenvalue (i.e. of multiplicity 1) of $A_{n\times n}$, then $\text{rank}(A-\lambda I) = n-1$.

(3)

Def: For an r-fold eigenvalue λ, r is called the algebraic multiplicity of λ.

Geometric multiplicity of $\lambda = n - \text{rank}(A-\lambda I)$
$= \#$ of zero rows in REF $(A-\lambda I)$

(*) $1 \leq$ geometric multiplicity $\leq$ algebraic multiplicity.

(*) λ is said to be regular when
geometric multiplicity = algebraic multiplicity.

Ex- $A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$

$\Psi_A(\lambda) = -(\lambda-2)^2(\lambda-8) = 0$

$\Rightarrow$ The eigenvalues of A are $2, 2,$ and 8.

Let $X = (x_1, x_2, x_3)^t$ be an eigenvector corresponding to 2

Then $(A - 2I)X = 0$

$$\Rightarrow \begin{pmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$\approx \begin{pmatrix} 4 & -2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \qquad \begin{matrix} R_2 = R_2 + \frac{1}{2}R_1 \\ R_3 = R_3 - \frac{1}{2}R_1 \end{matrix}$$

[$\because$ geometric multiplicity of 2 is two.

$\Rightarrow \quad 4x_1 - 2x_2 + 2x_3 = 0$

$\Rightarrow \quad 2x_1 - x_2 + x_3 = 0$ (Here x_2, x_3 are free variables)

Choose $x_2 = c \in \mathbb{R}$ and $x_3 = t \in \mathbb{R}$ then $x_1 = \frac{c}{2} - \frac{t}{2}$

$$\therefore X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} \frac{1}{2}c - \frac{1}{2}t \\ c \\ t \end{pmatrix}$$

To get two independent solutions for X, X_1 and X_2 put $c=1, t=0$ and $c=0, t=1$ $\Rightarrow X_1 = \begin{pmatrix} \frac{1}{2} \\ 1 \\ 0 \end{pmatrix}$ $X_2 = \begin{pmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{pmatrix}$

Similarly for eigenvalue 8 we have

$$(A - 8I)\,x = 0$$

$$\Rightarrow \begin{pmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$\sim \begin{pmatrix} -2 & -2 & 2 \\ 0 & -3 & -3 \\ 0 & -3 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \qquad \begin{matrix} R_2 = R_2 - R_1 \\ R_3 = R_3 + R_1 \end{matrix}$$

$$\sim \begin{pmatrix} 1 & 1 & -1 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \qquad \begin{matrix} R_1 = -\frac{1}{2} R_1 \\ R_3 = R_3 - R_2 \end{matrix}$$

[geometric multiplicity of 8 is one]

$$\Rightarrow \begin{cases} x_1 + x_2 - x_3 = 0 \\ -3x_2 - 3x_3 = 0 \end{cases}$$

only x_3 is free variable

Choose $x_3 = t \neq 0 \in \mathbb{R} \Rightarrow x_2 = -t \quad x_1 = 2t$

$t \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$ are eigenvectors corresponding to 8.

Note here both the eigenvalues algebraic multiplicity = geometric multiplicity. Thus both the eigenvalues are regular.